A DIGITAL APPROACH TO FISH IDENTIFICATION

Empowering Consumers with Knowledge of Commercially Valuable Species



Software Development Template · Species Data & App Concept

How the App Works

Three simple steps take a consumer from a raw photo to trusted species knowledge.

01



Scan

User photographs a fish at the market, harbor, or dinner table using the in-app camera.

→

02



Identify

The model matches the image against a commercial-species reference library and returns a confident match.

→

03



Learn

The app displays taxonomy, ecology, nutrition, and market price for the identified species.

Front Page — Scan & Identify

The opening screen a consumer sees: point the camera at a fish, and the software returns an instant identification.

Live viewfinder

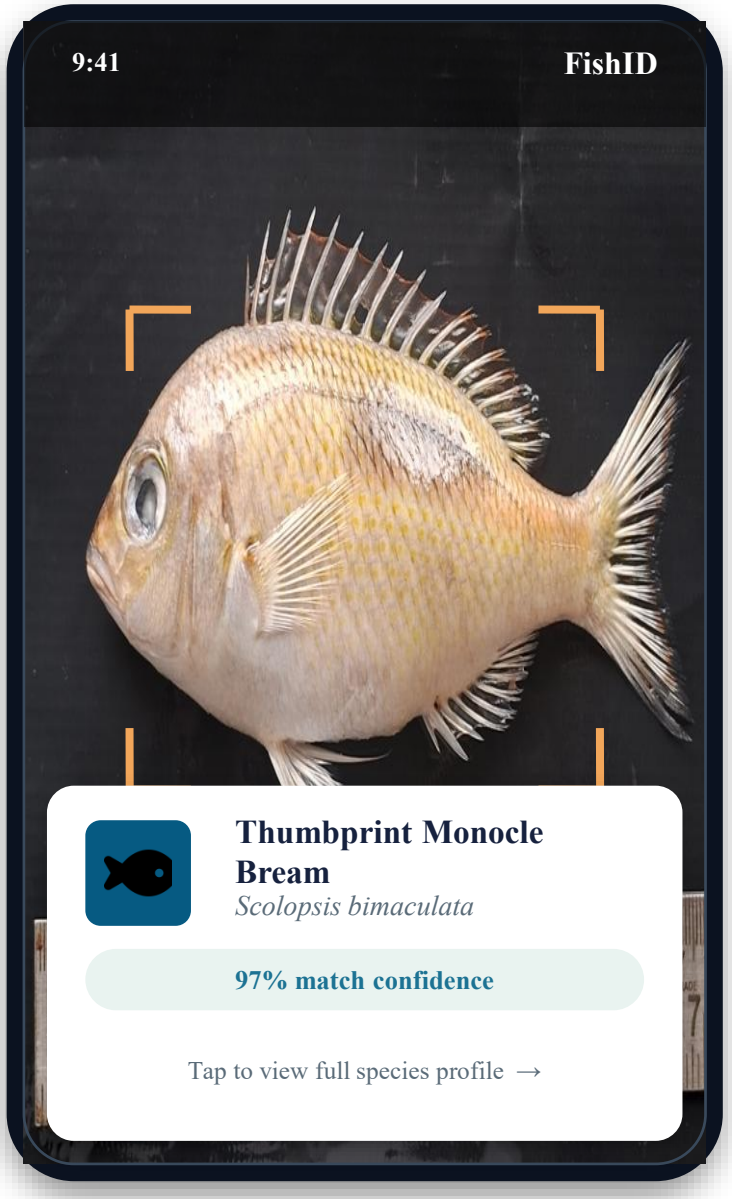
Real-time framing guide centers the fish for a clean capture.

Instant match

On-device or cloud inference returns a result in seconds.

Confidence score

Species name shown with a match-confidence percentage.



EXAMPLE SPECIES RECORD



This is the layer of detail each identified species unlocks for the user.

Species Profile

COMMON NAME

Thumbprint Monocle Bream

SCIENTIFIC NAME

Scolopsis bimaculata

SCIENTIFIC NAME - TAMIL

ஸ்கோலோப்சிஸ் பைமேக்குலேட்டா

VERNACULAR NAME

குண்டல் மீன்

This deck walks through the data fields a full species record contains — the same structure the app surfaces to consumers after each scan.

Morphological Attributes

Field identification notes used to visually confirm a match.

KEY FIELD MARKS

- Blue stripe joining the eyes
- 1–2 brownish blotches on the upper body
- Large suborbital spine present
- Antrorse (forward-directed) suborbital spine absent
- Head scales reach forward to the posterior nostrils
- Lower limb of preopercle scaly
- Pelvic fins long, reaching between anus and anal-fin origin; axillary scale present
- Colour: upper body grey, whitish below, with a blue stripe joining the eyes
- Body shape fusiform / normal; cross-section oval

MERISTIC COUNTS

10	9
Dorsal spines	Dorsal soft rays
3	**7**
Anal spines	Anal soft rays

Ecological Characteristics

Where the species lives, how it feeds, and its conservation status.



DISTRIBUTION

East and West coasts of India



HABITAT

Marine, reef-associated



FEEDING GUILD

Carnivore



VERTICAL GUILD

Demersal



DEPTH RANGE

0 – 60 m

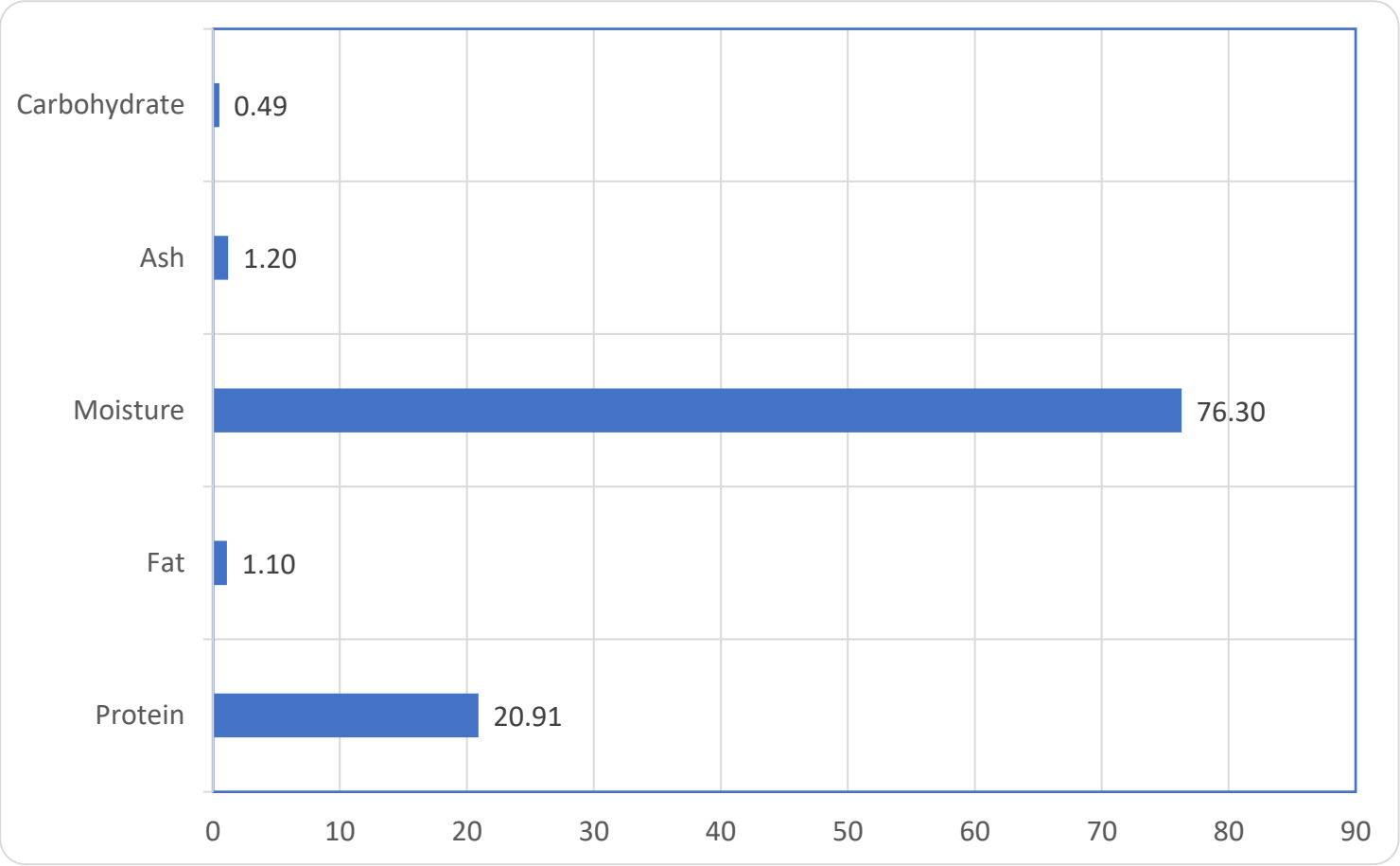


IUCN STATUS

Least Concern (LC)

Nutritional Composition

Composition per 100 g of edible portion, shown to consumers alongside each species.



PER 100 g

Protein	**20.91 g**
Fat	**1.10 g**
Moisture	**76.30 g**
Ash	**1.20 g**
Carbohydrate	**0.49 g**

Average Market Price

Commercial value shown to consumers to support purchasing decisions.



₹400

per kilogram

Prices are drawn from local market surveys and update per region and season — the app should treat this field as refreshable, not static

Species Record — Data Schema

What this template asks the software to store and surface for every commercial species.

Reference Image(s)

One or more labeled photos used for visual matching

Identification

Common name, scientific name, and regional-language name

Morphology

Field marks, meristic counts, body shape and colouration

Ecology

Distribution, habitat, feeding guild, depth range, IUCN status

Nutrition

Protein, fat, moisture, ash, carbohydrate per 100 g

Market Data

Average price per kilogram, refreshable by region/season

Metadata

Source, confidence threshold, last-updated timestamp

Contents

1. Species Profile
    - COMMON NAME
    - SCIENTIFIC NAME
    - SCIENTIFIC NAME – TAMIL
    - VERNACULAR NAME
2. Species Picture
3. Morphological Attribute
4. Ecological Characteristics
    - DISTRIBUTION
    - HABITAT
    - FEEDING GUILD
    - VERTICAL GUILD
    - DEPTH RANGE
    - IUCN STATUS
5. Nutritional Composition
6. Average Market Price



Let's build it together

This template shows the data and screens the app needs for one species — ready to scale across every commercially valuable fish in the catalogue.